

66. Fetal heart activity (The pulsating heart of the embryo) can be visualized using Doppler ultrasonography starting from _____ when counted from the LMP.
 A) 5 weeks
 B) 7 weeks
 C) 9 weeks
 D) 10 weeks
67. The fetal period is characterized by all of the following, EXCEPT:
 A) It begins 11 weeks after the LMP and ends at birth.
 B) Characterized by rapid body growth and differentiation of tissues and organ systems.
 C) Relative slowing of head growth compared with that of the rest of the body.
 D) The fetus is highly vulnerable to teratogens as compared to the embryonic period.
 E) None of the above
68. Which one of the following changes/events occur during the first 4 weeks of the fetal period?
 A) Growth of the head slows down but it is still disproportionately large compared with the rest of the body.
 B) The fetal movement can easily be felt by the mother.
 C) The bone becomes the major site of erythropoiesis.
 D) Limb movements are visible during ultrasound examination.
 E) All of the above
69. The primitive streak first appears at the beginning of the _____ week.
 A) First
 B) Second
 C) Third
 D) Fourth
 E) fifth
70. The most common site of ectopic pregnancy is:
 A) internal os of the uterus
 B) mesentery
 C) ovary
 D) uterine tube
 E) other
71. With the light microscope, the zona pellucida appears as a translucent membrane surrounding the:
 A) primary oocyte
 B) zygote
 C) morula
 D) very early blastocyst
 E) all of the above are correct
72. The amniotic cavity develops:
 A) on the tenth day
 B) within the outer cell mass
 C) within the inner cell mass near the cytotrophoblast
 D) in extraembryonic mesoderm
 E) none of the above
73. Cells from the primitive streak DO NOT become:
 A) Endoderm
 B) intermediate mesoderm
 C) paraxial mesoderm
 D) lateral plate mesoderm
 E) amnioblasts

10. Which one of the following muscles forms the floor of axillary armpit of the left arm?
 A. Internal oblique
 B. External oblique
 C. Latissimus dorsi
 D. Trapezius
 E. Serratus anterior
11. Which of the following muscles of the back is not innervated by the dorsal root of spinal nerves?
 A. Iliocostalis lumborum
 B. Longissimus thoracis
 C. Multifidus
 D. Serratus capitis
 E. Serratus posterior superior
12. With respect to their articular surfaces, one of the following joint is the strongest compared to the others
 A. Carpometacarpal joint of pollicis
 B. Humeroulnar joint
 C. Sternoclavicular joint
 D. Radiocarpal joint
 E. Proximal radioulnar joint
13. If extension at the elbow appeared normal, but supination of the forearm is weak, this indicates damage to
 A. All posterior divisions of the brachial plexus
 B. Radial nerve at the middle third of the humerus
 C. Posterior cord of the brachial plexus in the axilla
 D. Musculocutaneous nerve in the axilla
 E. Radial nerve injury near the head of the radius
14. If you slide a piece of paper between a patient's fingers and asked the patient to hold onto the paper when you try to pull it from the fingers, and the patient is unable to perform this action, which nerve is suspected to be nonfunctional?
 A. Deep branch radial nerve
 B. Superficial branch of ulnar nerve
 C. Recurrent branch of median nerve
 D. Deep branch of ulnar nerve
 E. Superficial branch of radial nerve
15. A "pulled elbow" in a young child results when ever
 A. There is wear and tear to the radial collateral ligament
 B. The radial head is dislodged from the annular ligament
 C. There is wear and tear to the Ulnar collateral ligament
 D. There is damage to insertion of tendon of Biceps brachii muscle
 E. There is fracture of the olecranon process
16. Which muscle does NOT cross the elbow joint and therefore has no action on the forearm
 A. Flexor pollicis longus
 B. Pronator teres
 C. Flexor carpi radialis
 D. Flexor carpi ulnaris
 E. Flexor digitorum superficialis

10. Which one of the following muscles forms the floor of axillary armpit of the left arm?
 A. Internal oblique
 B. External oblique
 C. Latissimus dorsi
 D. Trapezius
 E. Rhomboid
11. Which of the following muscles of the back is not innervated by the dorsal root of spinal nerves?
 A. Iliocostalis lumborum
 B. Longissimus thoracis
 C. Multifidus
 D. Serratus capitis
 E. Serratus posterior superior
12. With respect to their articular surfaces, one of the following joint is the strongest compared to the others
 A. Carpometacarpal joint of pollicis
 B. Humeroulnar joint
 C. Sternoclavicular joint
 D. Radiocarpal joint
 E. Proximal radioulnar joint
13. If extension at the elbow appeared normal, but supination of the forearm is weak, this indicates damage to
 A. All posterior divisions of the brachial plexus
 B. Radial nerve at the middle third of the humerus
 C. Posterior cord of the brachial plexus in the axilla
 D. Musculocutaneous nerve in the axilla
 E. Radial nerve injury near the head of the radius
14. If you slide a piece of paper between a patient's fingers and asked the patient to hold onto the paper when you try to pull it from the fingers, and the patient is unable to perform this action, which nerve is suspected to be nonfunctional?
 A. Deep branch radial nerve
 B. Superficial branch of ulnar nerve
 C. Recurrent branch of median nerve
 D. Deep branch of ulnar nerve
 E. Superficial branch of radial nerve
15. A "pulled elbow" in a young child results when ever
 A. There is wear and tear to the radial collateral ligament
 B. The radial head is dislodged from the annular ligament
 C. There is wear and tear to the Ulnar collateral ligament
 D. There is damage to insertion of tendon of Biceps brachii muscle
 E. There is fracture of the olecranon process
16. Which muscle does NOT cross the elbow joint and therefore has no action on the forearm
 A. Flexor pollicis longus
 B. Pronator teres
 C. Flexor carpi radialis
 D. Flexor carpi ulnaris
 E. Flexor digitorum superficialis

17. Which of the following is true in respect to the acromioclavicular joint?
- It is a saddle type synovial joint.
 - It is strengthened by the coracoclavicular and transverse humeral ligaments.
 - It is supplied by the lateral thoracic artery.
 - It is innervated by the nerve to the subscapularis.
18. Which of the following joints is paired correctly with its type?
- Proximal radioulnar joint... condyloid type of synovial joint
 - Distal radioulnar joint... pivot type of synovial joint
 - Interphalangeal joint... plane type of synovial joint
 - Atlantoaxial joint... hinge type of synovial joint
 - Atlantooccipital joint... condyloid type of synovial joint
19. An upper extremity flexion injury due to a fall to the ground and landing on the shoulder with the arm and humerus widely separated results all the following except
- Erb-Duchenne palsy
 - Disruption of C5-C6
 - Abductor shoulder
 - Medially rotated arm
 - Extended elbow
20. A lower extremity flexion injury in a patient who tried to catch and grab a tree by falling may result in all the following except
- The patient exhibits a "claw hand"
 - Klumpke's palsy
 - Damage to C8-T1
 - The patient exhibits
 - Motor loss to the hand
21. While you are trying to make a fist, your index and middle fingers remain extended. What nerve do you expect to be injured and where?
- Deep branch radial nerve at the wrist
 - Ulnar nerve at the elbow
 - Median nerve in the carpal tunnel
 - Posterior radial nerve
 - Recurrent nerve
22. The shoulder is most often dislocated in which direction?
- Anterior
 - Inferior
 - Posterior
 - Superior
 - Superior and medial
23. Which of the following is true in respect to the elbow joint?
- It is a plane type of synovial joint.
 - It is strengthened by the radial and ulnar cruciate
 - It is supplied by the cephalic and basilic arterie
 - It is innervated by the median and axillary ne
 - It is surrounded by the intratendinous olecr
- bursa, and the subcutaneous olecranon bursa

17. Which of the following is true in respect to the acromioclavicular joint?
- It is a saddle type synovial joint.
 - It is strengthened by the coracoclavicular and transverse humeral ligaments.
 - It is supplied by the lateral thoracic artery.
 - It is innervated by the nerve to the subscapularis.
18. Which of the following joints is paired correctly with its type?
- Proximal radioulnar joint... condyloid type of synovial joint
 - Atlantoaxial joint... pivot type of synovial joint
 - Interphalangeal joint... plane type of synovial joint
 - Atlantohippocentral joint... hinge type of synovial joint
 - Atlantoaxial joint... condyloid type of synovial joint
19. An upper extremity flexion injury due to a fall to the ground and landing on the shoulder with the arm and humerus widely separated results all the following except
- Erb-Duchenne palsy
 - Disruption of C5-C6
 - Abductor shoulder
 - Medially rotated arm
 - Extended elbow
20. A lower extremity flexion injury in a patient who tried to catch and grab a tree branch while falling may result in all the following except
- The patient exhibits a "claw hand"
 - Kumpke's gait
 - Damage to C8-T1
 - The patient exhibits
 - Motor loss to the hand
21. While you are trying to make a fist, your index and middle fingers remain extended. What nerve do you expect to be injured and where?
- Deep branch radial nerve at the wrist
 - Ulnar nerve at the elbow
 - Median nerve in the carpal tunnel
 - Posterior radial nerve
 - Recurrent nerve
22. The shoulder is most often dislocated in which direction?
- Anterior
 - Inferior
 - Posterior
 - Superior
 - Superior
23. Which of the following is true in respect to the elbow joint?
- It is a plane type of synovial joint.
 - It is strengthened by the radial and ulnar cruciate
 - It is supplied by the cephalic and basilic arteries
 - It is innervated by the median and axillary nerves
 - It is surrounded by the intratendinous olecranon bursa, and the subcutaneous olecranon bursa

17. Which of the following is correct in respect to the acromioclavicular joint?
- It is a saddle type synovial joint.
 - It is strengthened by the coracoclavicular and transverse humeral ligaments.
 - It is a ball and socket joint referred to as a "separated shoulder."
 - It is supplied by the lateral thoracic artery.
 - It is innervated by the nerve to the subscapularis.
18. Which of the following joints is paired correctly with its type?
- Proximal radioulnar joint... condyloid type of synovial joint
 - Distal radioulnar joint... pivot type of synovial joint
 - Interphalangeal joint... plane type of synovial joint
 - Atlantoaxial joint... hinge type of synovial joint
 - Atlantooccipital joint... condyloid type of synovial joint
19. An upper extremity flexion injury due to a fall to the ground and landing on the shoulder with the arm and humerus widely separated results all the following except
- Erb-Duchenne palsy
 - Disruption of C5-C6
 - Absent shoulder
 - Medially rotated arm
 - Extended elbow
20. A lower extremity flexion injury in a patient who tried to catch and grab a tree by falling may result in all the following except
- The patient exhibits a "claw hand"
 - Klumpke's palsy
 - Damage to C8-T1
 - The patient exhibits
 - Motor loss to the hand
21. While you are trying to make a fist, your index and middle fingers remain extended. What nerve do you expect to be injured and where?
- Deep branch radial nerve at the wrist
 - Ulnar nerve at the elbow
 - Median nerve in the carpal tunnel
 - Posterior radial nerve
 - Recurrent nerve
22. The shoulder is most often dislocated in which direction?
- Anterior
 - Inferior
 - Posterior
 - Superior
 - Superior and medial
23. Which of the following is true in respect to the elbow joint?
- It is a plane type of synovial joint.
 - It is strengthened by the radial and ulnar cruciate
 - It is supplied by the cephalic and basilic arteries
 - It is innervated by the median and axillary nerves
 - It is surrounded by the intratendinous olecranon bursa, and the subcutaneous olecranon bursa

59. Sacrococcygeal Teratoma
- A) Remnants of the primitive streak.
 - B) A benign tumors contain tissues derived from all three germ layers.
 - C) Most affected infants (80%) are female
 - D) All of the above
 - E) None of the above
60. The notochord
- A) originate as a solid cord of cells from the cranial end of primitive node
 - B) It persists in the adult only as the nucleus pulposus of the intervertebral discs.
 - C) It induces the overlying ectoderm to thicken and form the neural plate
 - D) Chordomas are tumors arising from vestigial remnants of notochordal tissue
 - E) All of the above
61. During _____ folding of the embryo Part of the yolk sac is incorporated in to the embryo and become the mid gut
- A) Head folding
 - B) Tail folding
 - C) Lateral folding
 - D) All
 - E) None
62. Which of the following structures does not turn under the ventral surface of the embryo during folding of the head?
- A) prochordal plate
 - B) heart
 - C) Connecting stalk
 - D) pericardial cavity
 - E) septum transversum
63. Before the folding of the embryo, the heart begins its development:
- A) Extraembryonically in a region anterior to the prochordal plate
 - B) Intraembryonically in a region ventral to the prochordal plate
 - C) Intraembryonically in a region caudal to the prochordal plate
 - D) In the intraembryonic mesoderm of the prochordal plate
 - E) None of the above
64. Not derivative of neural crest cell
- A) Dorsal root ganglion of spinal nerve
 - B) Neurolemma sheaths of peripheral nerves
 - C) Adrenal medulla
 - D) Autonomic ganglion
 - E) None of the above
65. All of the following epithelial linings/cells are derived from endoderm, except-
- A) Epithelial lining of the GIT, respiratory tract
 - B) Epithelial cells of the tonsils, thyroid gland
 - C) Neuroepithelium of the sense organs
 - D) Epithelial lining of eustachian tube
 - E) None of the above

10. Which one of the following muscles forms the floor of lumbar triangle of the back region?
A. Internal oblique
B. External oblique
C. Latissimus dorsi
D. Transverse abdominis
E. Quadratus lumborum
11. Which of the following muscles of the back is not innervated by the dorsal rami of spinal nerves?
A. Iliocostalis lumborum
B. Longissimus thoracis
C. Multifidus
D. Splenius capitis
E. Serratus posterior superior
12. With respect to their articular surfaces, one of the following joint is the strongest compared to the others
A. Carpometacarpal joint of pollicis
B. Humeroulnar joint
C. Sternoclavicular joint
D. Radiocarpal joint
E. Proximal radioulnar joint
13. If extension at the elbow appeared normal, but supination of the forearm is weak this indicates damage to
A. All posterior divisions of the brachial plexus
B. Radial nerve at the middle third of the humerus
C. Posterior cord of the brachial plexus in the axilla
D. Musculocutaneous nerve in the axilla
E. Radial nerve injury near the head of the radius
14. If you slide a piece of paper between a patient's fingers and asked the patient to hold onto the paper when you try to pull it from the fingers, and the patient is unable to perform this action, which nerve is suspected to be nonfunctional?
A. Deep branch of radial nerve
B. Superficial branch of ulnar nerve
C. Recurrent branch of median nerve
D. Deep branch of ulnar nerve
E. Superficial branch of radial nerve
15. A "pulled elbow" in a young child results when ever
A. There is wear and tear to the radial collateral ligament
B. The radial head is dislodged from the annular ligament
C. There is wear and tear to the Ulnar collateral ligament
D. There is damage to insertion of tendon of Biceps brachii muscle
E. There is fracture of the olecranon process
16. Which muscle does NOT cross the elbow joint and therefore has no action on the forearm
A. Flexor pollicis longus
B. Pronator teres
C. Flexor carpi radialis
D. Flexor carpi ulnaris
E. Flexor digitorum superficialis

Which of the following is true in respect to the acromioclavicular joint?

- A. It is a saddle-type synovial joint.
- B. It is strengthened by the coracoclavicular and transverse humeral ligaments.
- C. When dislocated, it is referred to as a "separated shoulder."
- D. It is supplied by the lateral thoracic artery.
- E. It is innervated by the nerve to the subclavius.

18. Which of the following joints is paired correctly with its type?

- A. Proximal and distal radioulnar joints ... condyloid type of synovial joints
- B. Radiocarpal joint ... pivot type of synovial joints
- C. Intercarpal joints ... plane type of synovial joints
- D. Metacarpophalangeal joints ... hinge type of synovial joints
- E. Interphalangeal joints ... condyloid type of synovial joints

19. An upper brachial plexus injury due to a fall to the ground and landing on the shoulder such that the neck and shoulder are widely separated results all the following except

- A. Erb-Duchenne palsy
- B. Damage to C5-C6
- C. Abducted shoulder
- D. Medially rotated arm
- E. Extended elbow

20. A lower brachial plexus injury in a patient who tried to catch and grab a tree branch while falling may result in all the following except

- A. The patient exhibits a "claw hand."
- B. Klumpke's palsy
- C. Damage to C8-T1
- D. The patient exhibit a "wrist drop"
- E. Motor loss to short muscles of the hand

21. While you are trying to make a fist, your index and middle fingers remained partially extended. What nerve do you expect to be injured and where?

- A. Deep branch radial nerve at the wrist
- B. Ulnar nerve at the elbow
- C. Median nerve in the carpal tunnel
- D. Posterior interosseous branch of radial nerve
- E. Recurrent branch of median nerve at the wrist

22. The shoulder is most often dislocated in which direction?

- A. Anterior
- B. Inferior
- C. Posterior

23. Which of the following is true in respect to the elbow joint?

- A. It is a plane type of synovial joint.
- B. It is strengthened by the radial and ulnar cruciate ligaments.
- C. It is supplied by the cephalic and basilic arteries.
- D. It is innervated by the median and axillary nerves.
- E. It is surrounded by the intratendinous olecranon bursa, the subcutaneous olecranon bursa, and the subcutaneous olecranon bursa.

17. Which of the following is false in respect to the acromioclavicular joint?
- It is a saddle type synovial joint.
 - It is strengthened by the coracohumeral and transverse humeral ligaments.
 - When dislocated, it is referred to as a "separated shoulder."
 - It is supplied by the lateral thoracic arteries.
 - It is innervated by the nerve in the subclavians.
18. Which of the following joints is paired correctly with its type?
- Proximal and distal radioulnar joints ... condyloid type of synovial joints
 - Radiocarpal joint ... pivot type of synovial joint
 - Intercarpal joints ... plane type of synovial joints
 - Metacarpophalangeal joints ... hinge type of synovial joints
 - Interphalangeal joints ... condyloid type of synovial joints
19. An upper brachial plexus injury due to a fall to the ground and landing on the shoulder such that the neck and shoulder are widely separated results all the following except
- Erb-Duchenne palsy
 - Damage to C5-C6
 - Abducted shoulder
 - Medially rotated arm
 - Extended elbow
20. A lower brachial plexus injury in a patient who tried to catch and grab a tree branch while falling may result in all the following except
- The patient exhibits a "claw hand."
 - Klumpke's palsy
 - Damage to C8-T1
 - The patient exhibit a "wrist drop"
 - Motor loss to short muscles of the hand
21. While you are trying to make a fist, your index and middle fingers remained partially extended. What nerve do you expect to be injured and where?
- Deep branch radial nerve at the wrist
 - Ulnar nerve at the elbow
 - Median nerve in the carpal tunnel
 - Posterior interosseous branch of radial nerve
 - Recurrent branch of median nerve at the wrist
22. The shoulder is most often dislocated in which direction?
- Anterior
 - Inferior
 - Posterior
 - Superolateral
 - Superomedial
23. Which of the following is true in respect to the elbow joint?
- It is a plane type of synovial joint.
 - It is strengthened by the radial and ulnar cruciate ligaments.
 - It is supplied by the cephalic and basilic arteries.
 - It is innervated by the median and axillary nerves.
 - It is surrounded by the intratendinous olecranon bursa, the subtendinous olecranon bursa, and the subcutaneous olecranon bursa.

43. Which of the following connective tissue cell originate locally from undifferentiated mesenchymal cell
- A. Fibroblast
 - B. Mast cell
 - C. Macrophages
 - D. Plasma cell
44. The cells of mononuclear phagocytic system in the liver tissue
- A. Dendritic cell
 - B. Macrophage
 - C. Kupffer cell
 - D. Langerhans cell
 - E. Dust cell
45. Types of protein fiber produced by the Mesenchymal type of embryonic connective tissue
- A. Collagen type I and III
 - B. Collagen type Type IV
 - C. Elastic fiber
 - D. Collagen type V
 - E. Reticular fiber
46. Anchoring protein in basement membrane formation found between the basal and reticular laminae is composed of
- A. Type VII collagen
 - B. Entactin
 - C. Laminin
 - D. Perlecan
 - E. Type VI collagen
47. The integral protein which allow the passage of small ions, molecules, and water across the plasma membrane in either direction (i.e., passive diffusion) of a cell.
- A. Channels
 - B. Receptor proteins
 - C. Linker proteins
 - D. Structural proteins
 - E. Pumps
48. The tissue composition of reticular lamina of the basement membrane is/are
- A. Collagen type IV
 - B. Collagen type I and II
 - C. Laminin and entactin
 - D. Perlecan
 - E. Anchoring protein
49. Which of the following fluid connective tissue cell lack azurophilic granules in the cytoplasm
- A. Neutrophils
 - B. Eosinophils
 - C. Basophils
 - D. Monocytes
 - E. None
50. The fibrous connective tissue underlying nearly all epithelia tissue is
- A. Loose connective tissue
 - B. Reticular Tissue
 - C. Adipose Tissue
 - D. Dens regular connective tissue
 - E. Dens irregular connective tissue
51. A type of tissue characterized with all cell resting on the basement membrane cell exposed to the surface is found in
- A. Esophagus
 - B. Primary bronchi
 - C. Urinary bladder
 - D. Uterus
 - E. Stomach

10. Which one of the following muscles forms the floor of lumbar triangle of the back region?
A. Internal oblique
B. External oblique
C. Latissimus dorsi
D. Transverse abdominis
E. Quadratus lumborum
11. Which of the following muscles of the back is not innervated by the dorsal rami of spinal nerves?
A. Iliocostalis lumborum
B. Longissimus thoracis
C. Multifidus
D. Splenius capitis
E. Serratus posterior superior
12. With respect to their articular surfaces, one of the following joint is the strongest compared to the others
A. Carpometacarpal joint of pollicis
B. Humeroulnar joint
C. Sternoclavicular joint
D. Radiocarpal joint
E. Proximal radioulnar joint
13. If extension at the elbow appeared normal, but supination of the forearm is weak this indicates damage to
A. All posterior divisions of the brachial plexus
B. Radial nerve at the middle third of the humerus
C. Posterior cord of the brachial plexus in the axilla
D. Musculocutaneous nerve in the axilla
E. Radial nerve injury near the head of the radius
14. If you slide a piece of paper between a patients fingers and asked the patient to hold onto the paper when you try to pull it from the fingers, and the patient is unable to perform this act which nerve is suspected to be nonfunctional?
A. Deep branch radial nerve
B. Superficial branch of ulnar nerve
C. Recurrent branch of median nerve
D. Deep branch of ulnar nerve
E. Superficial branch of radial nerve
15. A "pulled elbow" in a young child results when ever
A. There is wear and tear to the radial collateral ligament
B. The radial head is dislodged from the annular ligament
C. There is wear and tear to the Ulnar collateral ligament
D. There is damage to insertion of tendon of Biceps brachii muscle
E. There is fracture of the olecranon process
16. Which muscle does NOT cross the elbow joint and therefore does not contribute to its stability?
A. Flexor pollicis longus
B. Pronator teres
C. Flexor carpi radialis
D. Flexor carpi ulnaris
E. Flexor digitorum superficialis

74. In the third week of human embryonic development:

- A) The amnion appears
- B) A bilaminar embryonic disc is formed
- C) The body stalk moves ventrally and joins with the yolk sac stalk to form the umbilical cord
- D) The neural plate is induced by the notochordal process
- E) All of the above

75. The following organs are derived from mesoderm EXCEPT:

- A) skeletal musculature
- B) musculature of blood vessels
- C) cardiac musculature
- D) suprarenal cortex
- E) suprarenal medulla

76. The most important region of the decidua for the nourishment of the conceptus is the decidua

- A) Frondosum
- B) Capsularis
- C) Parietalis
- D) Basalis
- E) laeve

77. The conduction of action potential is influenced by:

- A. The diameter of axon
- B. Thickness of Myelin
- C. Temperature
- D. All of them

78. Which of the following events in postsynaptic neuronal membrane underlie inhibitory post synaptic potential? Opening of:

- A. Na channels
- B. K channels
- C. Ca channels
- D. All of the above

79. Which of the following statement is false about local potential?

- A. They are graded response
- B. Produced by threshold stimulus
- C. They are none-propagated response
- D. They may depolarize or hyperpolarize

80. A stronger than normal stimulus can cause excitation of nerve or muscle during?

- A. Absolute refractory period
- B. Over shoot
- C. Relative refractory period
- D. Threshold depolarization

81. The repolarization of action potential in nerve fiber is primarily due to?

- A. Na- K ATPase
- B. K efflux via voltage gated K channels
- C. Na influx via voltage gated Na channels
- D. Ca efflux via voltage gated Ca channels

82. Which one of the following is the commonest synapses in the CNS?

- A. Axosomatic synapses
- B. Axoaxonic synapses
- C. Axodendritic synapses
- D. Dendrodendritic synapses

83. The inhibitory amino acid neurotransmitter in the CNS are?

- A. Acetyl choline and Glutamate
- B. Aspartate and Glutamate
- C. GABA and Glycine
- D. Aspartate and Glycine

84. The resting potential of myelinated nerve fiber is primarily dependent on the concentration of which of the following ions?

- A. Na⁺
- B. K⁺
- C. Cl⁻
- D. Ca⁺⁺

85. Which of the following is the cause of resting membrane potential?
- An outward diffusion of K^+ through K^+ leak channels
 - Protein anions within the cell
 - The Na^+-K^+ pump
 - All of them
86. All of the following are not true about synapses except:
- Presence of synaptic cleft / delay in electrical synapses
 - Chemical synapses are always excitatory
 - Chemical synapses use neurotransmitters / cause bidirectional transmission
 - Electrical synapses cause bidirectional and quick transmission
87. Identify the incorrect statement about absolute refractory period:
- The period of time in which Na^+ channel gate will not open
 - It is period during which further stimulus does not cause another action potential
 - The period of time in which Na^+ channel gate will not re-open
 - Second action potential can never be elicited even with 500% threshold stimulus
88. Which types of channels are sensitive to open and close to respond to deformation in plasma membrane?
- Mechanically gated channels
 - Voltage-gated channels
 - Chemically gated channels
 - Leakage channels
89. Muscarinic receptor stimulation results in which of the following physiological changes?
- Cardiac acceleration
 - Bronchodilation
 - Adrenaline secretion
 - Gastrointestinal secretion
 - Skeletal muscle contraction
90. The stress reaction of human body is manifested by
- Stimulation of the vagus nerve to stomach
 - Release of adrenaline
 - Inhibition of sympathetic nerve to skeletal muscles
 - Bronchoconstriction
91. One of the following is does not contain category of Efferent nerve
- Preganglionic sympathetic nerve
 - Postganglionic sympathetic nerve
 - Preganglionic parasympathetic nerves
 - Postganglionic parasympathetic nerves
92. Which one of the following is not the characteristic of action potential?
- Action potentials are produced when local potential reaches threshold
 - Action potentials are all-or-none
 - Repolarization is a result of inflow of K^+ ion
 - Stimulus strength determines the frequency of action potential

17. Which of the following is true in respect to the acromioclavicular joint?
- A. It is a saddle-type synovial joint.
 - B. It is strengthened by the coracohumeral and transverse humeral ligaments.
 - C. When dislocated, it is referred to as a "separated shoulder."
 - D. It is supplied by the lateral thoracic arteries.
 - E. It is innervated by the nerve to the subclavius.
18. Which of the following joints is paired correctly with its type?
- A. Proximal and distal radioulnar joints ... condyloid type of synovial joints
 - B. Radiocarpal joint ... pivot type of synovial joint
 - C. Intercarpal joints ... plane type of synovial joints
 - D. Metacarpophalangeal joints ... hinge type of synovial joints
 - E. Interphalangeal joints ... condyloid type of synovial joints
19. An upper brachial plexus injury due to a fall to the ground and landing on the shoulder such that the neck and shoulder are widely separated results in all the following except
- A. Erb-Duchenne palsy
 - B. Damage to C5-C6
 - C. Abducted shoulder
 - D. Medially rotated arm
 - E. Extended elbow
20. A lower brachial plexus injury in a patient who tried to catch and grab a tree branch while falling may result in all the following except
- A. The patient exhibits a "claw hand."
 - B. Klumpke's palsy
 - C. Damage to C8-T1
 - D. The patient exhibit a "wrist drop"
 - E. Motor loss to short muscles of the hand
21. While you are trying to make a fist, your index and middle fingers remained partially extended. What nerve do you expect to be injured and where?
- A. Deep branch radial nerve at the wrist
 - B. Ulnar nerve at the elbow
 - C. Median nerve in the carpal tunnel
 - D. Posterior interosseous branch of radial nerve
 - E. Recurrent branch of median nerve at the wrist
22. The shoulder is most often dislocated in which direction?
- A. Anterior
 - B. Inferior
 - C. Posterior
 - D. Superolateral
 - E. Superomedial
23. Which of the following is true in respect to the elbow joint?
- A. It is a plane type of synovial joint.
 - B. It is strengthened by the radial and ulnar cruciate ligaments.
 - C. It is supplied by the cephalic and basilic arteries.
 - D. It is innervated by the median and axillary nerves.
 - E. It is surrounded by the intratendinous olecranon bursa, the subtendinous olecranon bursa, and the subcutaneous olecranon bursa.

17. Which of the following is true in respect to the acromioclavicular joint?
- A. It is a saddle-type synovial joint.
 - B. It is strengthened by the coracoclavicular and transverse humeral ligaments.
 - C. When dislocated, it is referred to as a "separated shoulder."
 - D. It is supplied by the lateral thoracic arteries.
 - E. It is innervated by the nerve to the subclavius.
18. Which of the following joints is paired correctly with its type?
- A. Proximal and distal radioulnar joints ... condyloid type of synovial joints
 - B. Radiocarpal joint ... pivot type of synovial joint
 - C. Intercarpal joints ... plane type of synovial joints
 - D. Metacarpophalangeal joints ... hinge type of synovial joints
 - E. Interphalangeal joints ... condyloid type of synovial joints
19. An upper brachial plexus injury due to a fall to the ground and landing on the shoulder such that the neck and shoulder are widely separated results in all the following except
- A. Erb-Duchenne palsy
 - B. Damage to C5-C6
 - C. Abducted shoulder
 - D. Medially rotated arm
 - E. Extended elbow
20. A lower brachial plexus injury in a patient who tried to catch and grab a tree branch while falling may result in all the following except
- A. The patient exhibits a "claw hand."
 - B. Klumpke's palsy
 - C. Damage to C8-T1
 - D. The patient exhibits a "wrist drop"
 - E. Motor loss to short muscles of the hand
21. While you are trying to make a fist, your index and middle fingers remained partially extended. What nerve do you expect to be injured and where?
- A. Deep branch radial nerve at the wrist
 - B. Ulnar nerve at the elbow
 - C. Median nerve in the carpal tunnel
 - D. Posterior interosseous branch of radial nerve
 - E. Recurrent branch of median nerve at the wrist
22. The shoulder is most often dislocated in which direction?
- A. Anterior
 - B. Inferior
 - C. Posterior
 - D. Superolateral
 - E. Superomedial
23. Which of the following is true in respect to the elbow joint?
- A. It is a plane type of synovial joint.
 - B. It is strengthened by the radial and ulnar cruciate ligaments.
 - C. It is supplied by the cephalic and basilic arteries.
 - D. It is innervated by the median and axillary nerves.
 - E. It is surrounded by the intratendinous olecranon bursa, the subtendinous olecranon bursa, and the subcutaneous olecranon bursa.

10. Which one of the following muscles forms the floor of lumbar triangle of the back region?
A. Internal oblique
B. External oblique
C. Latissimus dorsi
D. Transverse abdominis
E. Quadratus lumborum
11. Which of the following muscles of the back is not innervated by the dorsal rami of spinal nerves?
A. Iliocostalis lumborum
B. Longissimus thoracis
C. Multifidus
D. Splenius capitis
E. Serratus posterior superior
12. With respect to their articular surfaces, one of the following joint is the strongest compared to the others
A. Carpometacarpal joint of pollicis
B. Humeroulnar joint
C. Sternoclavicular joint
D. Radiocarpal joint
E. Proximal radioulnar joint
13. If extension at the elbow appeared normal, but supination of the forearm is weak this indicates damage to
A. All posterior divisions of the brachial plexus
B. Radial nerve at the middle third of the humerus
C. Posterior cord of the brachial plexus in the axilla
D. Musculocutaneous nerve in the axilla
E. Radial nerve injury near the head of the radius
14. If you slide a piece of paper between a patient's fingers and asked the patient to hold onto the paper when you try to pull it from the fingers, and the patient is unable to perform this action which nerve is suspected to be nonfunctional?
A. Deep branch radial nerve
B. Superficial branch of ulnar nerve
C. Recurrent branch of median nerve
D. Deep branch of ulnar nerve
E. Superficial branch of radial nerve
15. A "pulled elbow" in a young child results when ever
A. There is wear and tear to the radial collateral ligament
B. The radial head is dislodged from the annular ligament
C. There is wear and tear to the Ulnar collateral ligament
D. There is damage to insertion of tendon of Biceps brachii muscle
E. There is fracture of the olecranon process
16. Which muscle does NOT cross the elbow joint and therefore has no action on the forearm?
A. Flexor pollicis longus
B. Pronator teres
C. Flexor carpi radialis
D. Flexor carpi ulnaris
E. Flexor digitorum superficialis

24. The synovial cavity of the glenohumeral joint communicates with the subdeltoid (subacromial) bursa after the rupture of the:
- A. Infraspinatus tendon
 - B. Middle glenohumeral ligament
 - C. Subacromial tendon
 - D. Supraspinatus tendon
 - E. Long head of the biceps brachii tendon
25. Ulnar nerve injury commonly occurs where the nerve passes behind the medial epicondyle of the humerus. Injury to the ulnar nerve may result in all of the following symptoms except
- A. loss of sensation on the medial surface of the palm and digits 4 and 5
 - B. loss of abduction of the wrist
 - C. loss of flexion at distal interphalangeal joints of digits 4 and 5
 - D. loss of abduction of digit 5
 - E. loss of adduction of the thumb
26. In "tennis elbow", the focal point of pain would most likely be near which palpable bony landmark?
- A. Coronoid process of ulna
 - B. Lateral epicondyle of humerus
 - C. Lateral supracondylar ridge of humerus
 - D. Medial epicondyle of humerus
 - E. Medial supracondylar ridge of humerus
27. A patient is severely limited in extension at the wrist joint after several months in a cast following a Colles fracture. Which joint would be especially important in therapy to regain full extension?
- A. Carpometacarpal
 - B. Distal radioulnar
 - C. Midcarpal
 - D. Radiocarpal
 - E. Proximal radioulnar
28. When falling on an outstretched hand, the most commonly dislocated carpal bone is the
- A. Scaphoid
 - B. Trapezoid
 - C. Lunate
 - D. Capitate
 - E. Hamate
29. Of the following muscles, which one is the strongest medial rotator of the arm?
- A. Coracobrachialis
 - B. Subscapularis
 - C. Infraspinatus
 - D. Supraspinatus
 - E. Teres minor
30. Pronounced weakness in supination of the forearm is most likely the result of damage which of the following nerves?
- A. Median nerve
 - B. Ulnar nerve
 - C. Musculocutaneous nerve
 - D. Anterior interosseous nerve
 - E. Superficial radial nerve

1. CHOOSE THE BEST ANSWER AND WRITE THE LETTER OF YOUR CHOICE ON SEPARATE ANSWER SHEET

1. Which of the following are functions of vertebral column?

A. Protect spinal cord and spinal nerves	C. Posture and locomotion
B. Support body weight	D. Placement of the head
	E. All of the above
2. Which one of the following bone is commonly related with odontoid process?

A. First cervical vertebral bone	D. First thoracic vertebral bone
B. Second cervical vertebral bone	E. Hyoid bone
C. Seventh cervical vertebral bone	
3. Which cervical vertebral bone is with the longest spinous process?

A. First	D. Fifth
B. Second	E. Eighth
C. Seventh	
4. An intervertebral foramina in the thoracic region is bounded posteriorly by:

A. Ligamentum flava	D. Intervertebral disc
B. Inferior vertebral notch	E. Vertebral body
C. Superior vertebral notch	
5. Lateral sacral crest is formed by fusion of which part of sacral vertebral bones?

A. Superior and inferior articular processes	C. Transverse processes
B. Bodies and intervertebral discs	D. Spinous processes
	E. Pedicles
6. Which one of the following back ligament is located most laterally?

A. Intertransverse ligament	D. Supraspinal ligament
B. Posterior longitudinal ligament	E. Interspinal ligament
C. Anterior longitudinal ligament	
7. Multifidus muscles of the back are classified into:

A. Spinotransversalis muscles	D. Short deep muscles of the neck
B. Sacrospinous muscles	E. Ventral neck muscles of the back
C. Transversospinalis muscles	
8. Latissimus dorsi muscle has attachment in the following point except:

A. Thoracolumbar fascia	D. Outer surface of inferior angle of scapula
B. Lower three or four ribs	E. Spinous process of upper ten thoracic vertebrae
C. Posterior part of iliac crest	
9. Which of the following statements concerning an intervertebral disc is not correct?

A. The discs are the thickest in the lumbar region
B. The atlantoaxial joint possesses no disc
C. During aging, the fluid within the nucleus pulposus is replaced by fibrocartilage
D. The nucleus pulposus is most likely to herniate in an anterolateral direction
E. The discs play a major role in the development of the curvatures of the vertebral column